

Engineering Drought Resilience Through Plant Architecture

Research Proposal

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INTRODUCTION & BACKGROUND: Droughts create unique complexities for agricultural production (Passioura, 2007). Water deficiencies reduce crop yield, increase irrigation demands, and complicate crop improvement efforts due to varying environmental and geographical conditions (Passioura, 2007). Engineering drought resilience could help alleviate these challenges by identifying quantifiable traits and characteristic combinations associated with improved performance under water-limited conditions. Water-use efficiency (WUE) describes the relationship between carbon gain and water usage; generally, higher WUE allows plants to achieve greater growth while using less water (Passioura, 2007). Improvements in WUE may help mitigate drought-related challenges while expanding cultivation potential into marginal lands. Additionally, understanding the traits associated with WUE may contribute to a broader understanding of plant hydraulics, crop breeding, and future predictive modeling efforts (Zhou et al., 2020).

RESEARCH QUESTION: Can drought tolerance and WUE be improved through changes in plant architecture, root systems, and hydraulic functions between tepary and common beans?

EXPLANATION OF HYPOTHESES: Native to the arid and hot regions of the southwestern United States and northern Mexico, tepary beans exhibit greater drought tolerance and higher water-use efficiency than common beans under water-limited conditions (Traub, 2015; Buckley et al., 2025). These characteristics suggest that inherent differences in plant architecture, hydraulic function, and growth regulation contribute to improved water acquisition, transport, and conservation. Because common beans are an agriculturally important staple crop yet remain relatively sensitive to drought stress, multiple approaches will be used to alter plant architecture and evaluate its relationship with drought resilience (Buckley et al., 2025). Removal of axillary (side) or apical (main) shoot meristems may alter branching patterns, while topical hormone applications in tepary beans may influence root and shoot development (Naeem et al., 2004; Pilet & Saugy, 1987). Likewise, different grafting combinations between tepary and common bean rootstocks and scions may produce distinct architectural and hydraulic outcomes by altering water uptake and transport (Fandrich, 1998; Gurusamy et al., 2010). These modifications are expected to influence transpiration, WUE, and drought performance. By quantifying architectural, anatomical, and physiological traits, this study seeks to identify relationships between plant structure, hydraulic function, and drought resilience (Buckley et al., 2025; Zhou et al., 2020).

METHODOLOGY

Hormonal Applications: Tepary beans (*Phaseolus acutifolius*) will receive indole-3-acetic acid (IAA), paclobutrazol (PAC), or water-control treatments to evaluate the effects of hormone-mediated growth regulation on plant architecture. Architectural traits including branching pattern, root development, and lateral root formation will be compared among treatments (Pilet & Saugy, 1987; Naeem et al., 2004). If significant architectural differences are observed, plants will be evaluated under drought stress.

Mechanical Removal of Apical and Axillary Meristems: Tepary beans will undergo apical or axillary meristem removal to alter apical dominance and branching patterns. Resulting architectural changes will be compared to untreated controls to assess their relationship with hydraulic function, WUE, and drought response.

Grafting: Common bean (*Phaseolus vulgaris*) and tepary bean will be grafted in both same-species and interspecific rootstock-scion combinations, and subsequently exposed to well-watered and drought-stress conditions (Fandrich, 1998; Gurusamy et al., 2010). Experimental grafts will be compared against non-grafted controls and same-species graft controls to determine how rootstock identity influences plant architecture, water transport, and drought tolerance (Fandrich, 1998; Gurusamy et al., 2010).

Physiological and anatomical measurements: Young and older leaves, defined as the youngest fully expanded leaf nearest the shoot apical meristem and the oldest leaf nearest the cotyledons, respectively, will be compared to evaluate age-related differences in transpiration and drought response. Plant architecture, including compactness, convex hull, leaf area, growth habit, branching pattern, growth rate, will be quantified using PhenoSight and PhenoCage (Yu et al., 2024). Physiological measurements will include stomatal conductance and transpiration using the SC-1 porometer, photosynthetic performance and photosystem II efficiency using PhotosynQ / PhenoSight, and plant water status and leaf water potential using PsyPro. Physiological and architectural measurements will be collected before, during, and after drought treatment to capture responses to water limitation. At the conclusion of each experiment, xylem and phloem structure, stomatal density, and leaf cross sections will be characterized using a Dinolite microscope and Rapid Anatomy Tools (RAT). These anatomical data will be incorporated into CPlantBox modeling to investigate relationships among vascular anatomy, hydraulic function, plant architecture, WUE, and drought tolerance (Zhou et al., 2020).

JUSTIFICATION: Tepary beans are adapted to arid environments, and thus may reveal how plant architecture and hydraulic function contribute to drought tolerance and WUE (Traub, 2015; Buckley et al., 2025). Through hormonal manipulation, meristem removal, and grafting, this study will generate architectural variation to examine how shoot branching patterns, rootstock contributions, and vascular organization influence plant water relations (Naeem et al., 2004; Pilet & Saugy, 1987; Fandrich, 1998; Gurusamy et al., 2010). By integrating phenotypic, physiological, and anatomical measurements with CPlantBox modeling, this work seeks to identify traits associated with improved hydraulic performance under drought stress (Yu et al., 2024; Zhou et al., 2020). Comparisons among tepary beans, common beans, and grafted combinations aim to provide insight into the mechanisms underlying drought adaptation (Traub, 2015; Buckley et al., 2025). Collected results will inform future breeding, engineering, and modeling efforts in beans and related legume crops.

ANTICIPATED RESULTS

1. Tepary beans are expected to exhibit greater drought tolerance and WUE compared to common beans due to differences in root architecture, hydraulic function, and water-use strategies (Traub, 2015; Buckley et al., 2025).
2. Reduced leaf area and compact growth habits are expected to correspond to improved drought tolerance (Hütsch & Schubert, 2021).
3. Plants grafted onto tepary rootstocks are expected to outperform plants grafted onto common bean rootstocks, non-grafted plant controls, and same-species graft controls, potentially through superior water transport (Fandrich, 1998; Gurusamy et al., 2010).
4. Younger leaves are expected to maintain higher water potential, stomatal conductance, and photosystem II efficiency than older leaves under drought stress, suggesting preferential hydraulic support of developing tissues.
5. IAA-treated plants are expected to perform better under drought stress than mock-treated controls, possibly due to greater lateral root development, which would improve water acquisition under drought stress (Pilet & Saugy, 1987; Naeem et al., 2004).
6. Paclobutrazol-treated plants are expected to exhibit reduced shoot growth and more compact architectures than mock-treated controls, potentially improving water conservation under drought stress (Hütsch & Schubert, 2021).
7. Removal of the apical meristem is expected to increase branching, while removal of axillary meristems is expected to reduce branching. These architectural differences are expected to affect water-use efficiency and drought tolerance, although the directionality of this response is difficult to predict.

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